

ChatGPT - Homework



L21_L22.pdf

PDF

You are given lecture slides and relevant textbook excerpts as the context. Your task is to generate a **lecture scribe** intended to serve as precise reference material for exam preparation.

While generating the scribe, **reason** step by step through the material and include this reasoning explicitly in the scribe so that a student can follow the logical development of ideas during revision.

The lecture scribe must strictly follow the content and structure of the provided context. **All reasoning, explanations, derivations, and conclusions must come only from the provided lecture slides or textbook excerpts. Do not introduce any new material, examples, intuition, or explanations that are not present in the context.**

When writing the scribe:

- Present **definitions, notation, assumptions, and conditions explicitly**.
- Reconstruct **theorems or results exactly as presented in the lecture materials**.
- Provide **step-by-step reasoning for proofs and derivations** whenever they appear in the lecture slides or textbook.
- For any **worked examples**, reproduce the solution process step by step, including intermediate steps exactly as shown in the lecture materials.
- Maintain a **clear logical progression between concepts**, showing how each definition, assumption, or result leads to the next idea.

The final output should:

- Be written in a **formal academic tone suitable for exam revision**.

- Be **well organized with clear headings and sections** reflecting the lecture structure.
- Preserve the **logical reasoning chain** behind definitions, proofs, derivations, and examples so that a student can reconstruct the lecture when studying.

Important constraints:

- Use **only the provided context** (lecture slides and relevant textbook content).
- **Do not invent content or fill gaps with external knowledge.**
- Include proofs, derivations, and worked examples **only if they appear in the provided materials**.

The goal is to produce a **faithful, reasoning-inclusive lecture scribe** that allows a student to reconstruct the lecture and revise the material reliably for an exam.

Here is a **faithful, reasoning-inclusive lecture scribe** constructed strictly from the provided slides (Lectures 21–22 on *Inequalities and Tail Bounds*).

Lecture 21–22: Inequalities and Tail Bounds

1. Tail of a Random Variable

Definition

The **tail** of a random variable X is defined as:

$$P(X > x)$$

This represents the probability that the random variable takes a value **greater than a given threshold**.

Motivation for Studying Tails

From the slides (pages 5–7), the importance of tails is motivated through real-world scenarios:

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

These correspond to **rare but extreme events**, captured by tail probabilities.

Key Idea

- Mean → captures **average behavior** (easy to compute)
 - Tail → captures **rare extreme events** (hard to compute)
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Why Are Tails Hard to Compute?

- Requires **summing many small probabilities**
- Leads to **complicated, long summations**

This difficulty motivates the need for **tail bounds** instead of exact computation.

2. Tail Examples

Example 1: Load Distribution Across Servers

Problem Statement

Suppose n jobs are distributed among n servers at random.

What is:

$$P(\text{a particular server gets } \geq k \text{ jobs})?$$

Reasoning from Slides

- Jobs are assigned randomly
- Most servers receive **few jobs**
- Occasionally, one server gets **many jobs**

This event represents a **tail event (rare overload)**.

Mathematical Formulation

$$X \sim \text{Binomial}(n, p)$$

$$P(X \geq k) = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$$

This is a **tail probability** of a **Binomial distribution**.

Example 2: Poisson Arrivals

Problem Statement

Jobs arrive according to a Poisson process with rate λ jobs/hour.

What is:

$$P(\text{at least } k \text{ jobs arrive in the first hour})?$$

Reasoning from Slides

- Jobs arrive randomly over time
- Most hours have **moderate arrivals**
- Sometimes, there is a **burst of arrivals**

This represents a **tail event** (sudden spike in load).

Mathematical Formulation

$$X \sim \text{Poisson}(\lambda)$$

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Again, this is a **tail probability** involving an infinite sum.

3. Why Tail Bounds?

Problem with Exact Computation

From page 14:

- Binomial tail:

$$P(X \geq k) = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$$

- Poisson tail:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

These are:

- **Complicated**
 - **Long summations**
 - **Difficult to compute exactly**
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Key Idea (Page 15)

Instead of computing exact values:

- Find an **upper bound**
- Easier to compute
- Still provides a **guarantee**

Exact: $P(X \geq k) = ?$

Bound: $P(X \geq k) \leq \text{something simple}$

Conclusion

Core principle:

Approximate safely instead of computing exactly.

4. Running Example (Key Idea)

We analyze all bounds using a common example.

Setup

- Flip a fair coin n times
- Let:

$X = \text{number of heads}$

Distribution

$$X \sim \text{Binomial}(n, \frac{1}{2})$$

Mean

$$E[X] = \frac{n}{2}$$

Tail Question

$$P\left(X \geq \frac{3n}{4}\right)$$

This represents **getting unusually many heads**, i.e., a **tail region**.

5. Markov's Inequality

Key Idea (Page 18)

- Uses **only the mean**
- Large values are **rare**

Example intuition:

- If average is 10, very few values can be 100+
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Statement of Markov's Inequality

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Proof (Step-by-Step from Slides)

Start with expectation:

$$E[X] = \sum_{x=0}^{\infty} x \cdot p_X(x)$$

Step 1: Restrict to large values

$$E[X] \geq \sum_{x=a}^{\infty} x \cdot p_X(x)$$

Step 2: Replace x by a (since $x \geq a$)

$$\geq \sum_{x=a}^{\infty} a \cdot p_X(x)$$

Step 3: Factor out a

$$= a \sum_{x=a}^{\infty} p_X(x)$$

Step 4: Recognize probability

$$= a \cdot P(X \geq a)$$

Final Inequality

$$\begin{aligned} E[X] &\geq a \cdot P(X \geq a) \\ \Rightarrow P(X \geq a) &\leq \frac{E[X]}{a} \end{aligned}$$

6. Markov's Inequality on Running Example

Given

- $X \sim \text{Binomial}(n, \frac{1}{2})$

- $E[X] = \frac{n}{2}$
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Apply Markov's Inequality

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{E[X]}{3n/4}$$

Substitute Mean

$$= \frac{n/2}{3n/4} = \frac{2}{3}$$

Conclusion

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{2}{3}$$

Observation (from Slide)

- This is a **terrible bound**
 - Reason: it **does not depend on n**
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7. Real-Life Interpretation

Example: Income Distribution

- Average income = 10,000

Using Markov's inequality:

- Very high incomes must be **rare**, since the average is limited

This demonstrates how **mean alone can bound extreme outcomes**.

8. Case Study Motivation

From the final slide:

How to predict the worst-case demands on a server?

This directly connects to:

- Tail probabilities
 - Bounding extreme events
 - Designing systems robust to rare spikes
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Final Summary

1. **Tail probability** measures extreme events: $P(X > x)$
 2. Exact computation is difficult due to **long summations**
 3. **Tail bounds** provide simpler, safe approximations
 4. **Markov's inequality**:
 - Uses only the mean
 - Provides a simple upper bound
 - Often **loose (weak)**
 5. Running example shows:
 - Tail events = rare
 - Bounds may lack tightness
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If you want, I can continue this scribe with **Chebyshev + Chernoff bounds** in the exact same strict format (those are where things actually get powerful—and where exam questions usually come from).